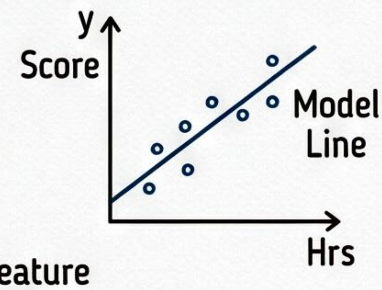
LINEAR REGRESSION

1. The Hypothesis Function (Best Fit Line)

Theory: We want to find the best straight line equation that fits our data to make predictions for new inputs.



Recall straight line:
$y = c + mx$ ⇒

$$\boxed{h_\theta(x) = \theta_0 + \theta_1 x_1}$$

- $h_\theta(x)$: Hypothesis fnx (our prediction model)
- θ_0: Bias / Intercept
- θ_1: 1st param / coeff (slope)
- x_1: 1st feature (input)

2. Cost Function

Theory: Measures how 'wrong' our model is. It calculates the difference between predictions and actual values. Our goal is to MINIMIZE this.

Prediction – Actual value

$$h_\theta(x^{(i)}) - y^{(i)}$$

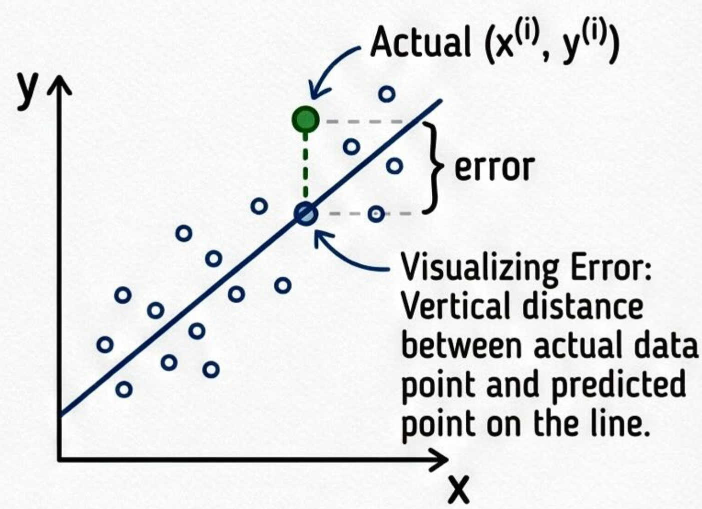
Types of error measurement:

- Squared error (common)
- Absolute error
- Log of error ...

⇒ Mean Squared Error (MSE)

$$\boxed{J(\theta) = \frac{1}{2m}\sum_{i=1}^{m}\left(h_\theta(x^{(i)}) - y^{(i)}\right)^2}$$

Why square it? To make all errors positive and penalize larger mistakes more. The '1/2' makes the derivative closer for gradient descent later.



3. Example Calculation (Continuous from page 2...)

Note: Let's apply the concepts to a simple dataset.

Data Points & Assumptions:

- Data: $(x^{(1)}, y^{(1)}) = (1,2)$, $(x^{(2)}, y^{(2)}) = (2,4)$, $(x^{(3)}, y^{(3)}) = (3,6)$
- Assume: $\theta_0 = 0$, $\theta_1 = 2$

1. Hypothesis Function & Predictions:

$$h_\theta(x) = \theta_0 + \theta_1 x = 0 + 2x = 2x$$

For $x=1$: $h_\theta(1) = 2(1) = 2$

For $x=2$: $h_\theta(2) = 2(2) = 4$

For $x=3$: $h_\theta(3) = 2(3) = 6$

↳ Notice: Predictions match actual y values perfectly.

2. Cost Function Calculation (MSE):

$$J(\theta) = \frac{1}{2m} \sum_{i=1}^{m} \left(h_\theta(x^{(i)}) - y^{(i)}\right)^2$$

Here, $m = 3$ (number of training examples).

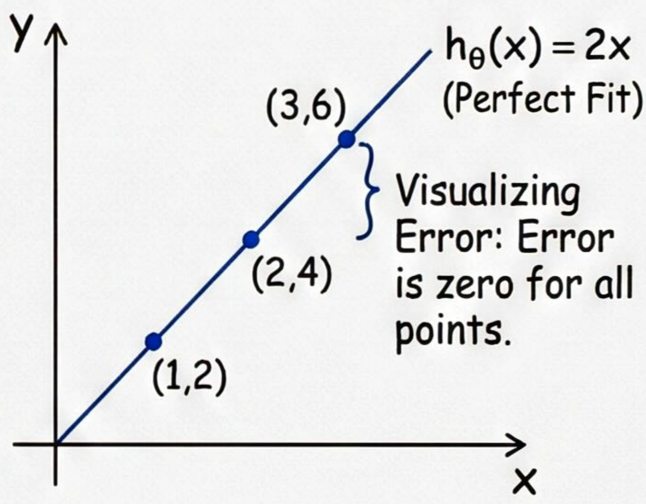
$$J(\theta) = \frac{1}{2 \times 3} \left[\left(h_\theta(1) - y^{(1)}\right)^2 + \left(h_\theta(2) - y^{(2)}\right)^2 + \left(h_\theta(3) - y^{(3)}\right)^2 \right]$$

$$J(\theta) = \frac{1}{6} \left[(2-2)^2 + (4-4)^2 + (6-6)^2 \right]$$

$$J(\theta) = \frac{1}{6} \left[0^2 + 0^2 + 0^2 \right] = \frac{1}{6} \times 0 = 0$$

Interpretation: $J(\theta) = 0$ means there is ZERO error. The model fits the data perfectly.

Plotting the Graph & Model Line



The model $h_\theta(x)=2x$ is the best fit for this data.

4. Visualizing the Cost Function $J(\theta)$

Objective: To understand how the cost $J(\theta)$ changes as we vary the parameter θ_1 (with $\theta_0 = 0$).

Recall

Dataset: $(1,2), (2,4), (3,6)$. $m=3$

Hypothesis: $h_\theta(x) = \theta_1 x$ (since $\theta_0 = 0$).

Cost: $J(\theta) = \frac{1}{2m}\sum_i (h_\theta(x^{(i)}) - y^{(i)})^2$

Calculating $J(\theta)$ for different θ_1 values:

- **Case 1**: $\theta_1 = 2$ (Best fit, from previous page)

  $h_\theta(x) = 2x$. Errors: $(2(1)-2)^2 = 0$, $(2(2)-4)^2 = 0$, $(2(3)-6)^2 = 0$.

  Errors: $[1+0] = [h_\theta(x^{(i)}+0] - y^{(i)})$

  $J(\theta) = \frac{1}{6}\cdot[0+0+0] = 0$. So, point $(\theta_1, J(\theta))$ is $(2, 0)$.

- **Case 2**: $\theta_1 = 1$

  $h_\theta(x) = 1x$. Errors: $(1(1)-2)^2 = (-1)^2 = 1$,
  $(1(2)-4)^2 = (-2)^2 = 4$,
  $(1(3)-6)^2 = (-3)^2 = 9$.

  $J(\theta) = \frac{1}{6}\cdot[1+4+9] = \frac{14}{6} \approx 2.34$.

  So, point $(\theta_1, J(\theta))$ is $(1, 2.34)$.

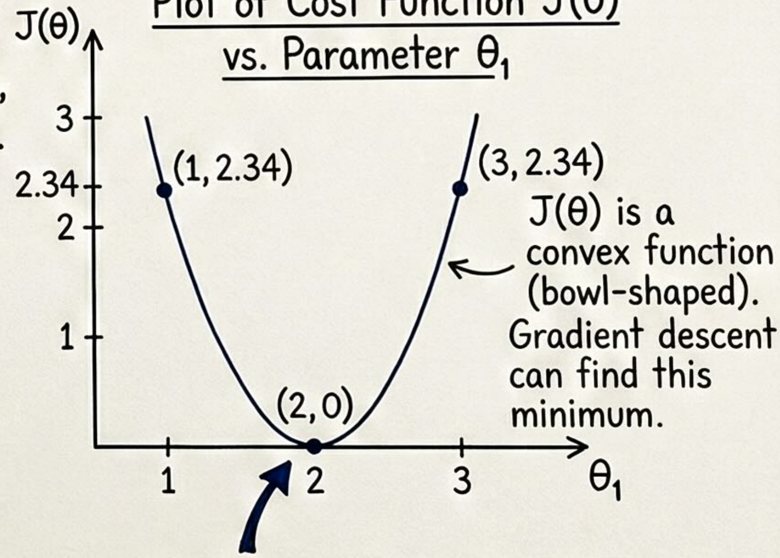
- **Case 3**: $\theta_1 = 3$

  $h_\theta(x) = 3x$. Errors: $(3(1)-2)^2 = (1)^2 = 1$,
  $(3(2)-4)^2 = (2)^2 = 4$,
  $(3(3)-6)^2 = (3)^2 = 9$.

  $J(\theta) = \frac{1}{6}\cdot[1+4+9] = \frac{14}{6} \approx 2.34$.

  So, point $(\theta_1, J(\theta))$ is $(3, 2.34)$.

Plot of Cost Function $J(\theta)$ vs. Parameter θ_1



Global Minimum: $\theta_1 = 2$ minimizes $J(\theta)$. This corresponds to the best fit line.

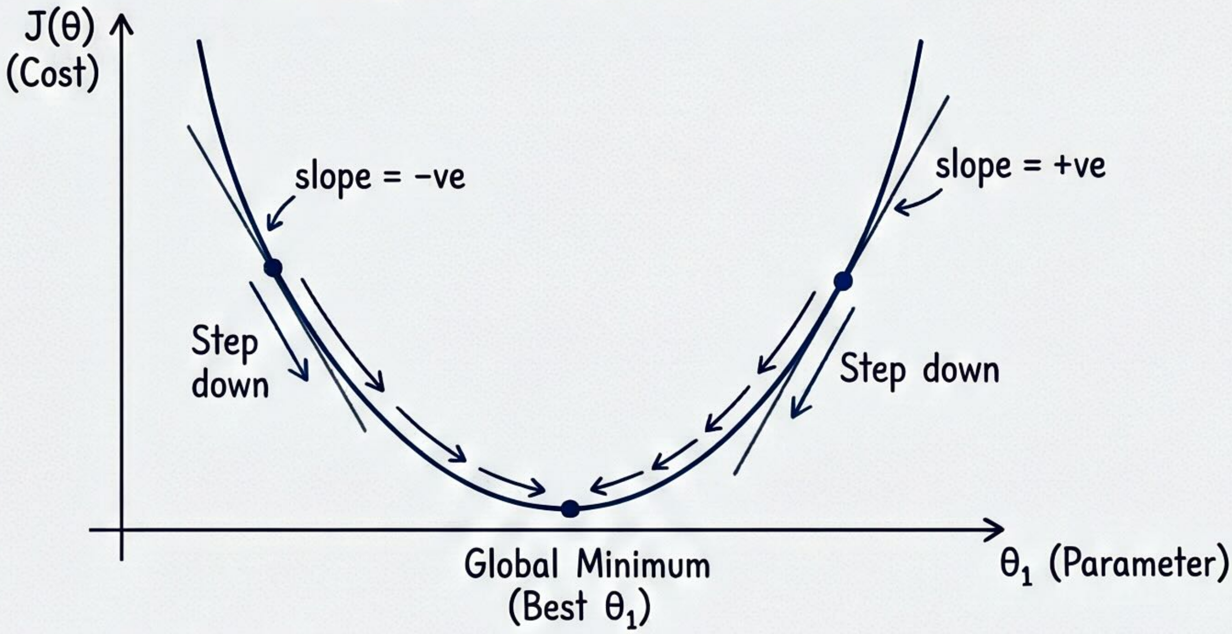
5. Gradient Descent

Theory & Intuition

Imagine you are on top of a hill (the cost function $J(\theta)$) and want to get to the very bottom (minimum cost) blindfolded. You feel the slope of the ground around you and take a step in the steepest downhill direction. You repeat this process until you reach the flattest point (the valley).

Goal: To find the values of θ_0 and θ_1 that minimize the cost function $J(\theta)$.

Visualizing Gradient Descent on $J(\theta)$



The Algorithm (Update Rule)

$$\boxed{\text{Repeat until convergence: } \left\{ \theta_j := \theta_j - \alpha \cdot \frac{\partial J(\theta)}{\partial \theta_j} \right\} \text{ (for } j=0 \text{ and } j=1\text{)}}$$

Parameter to update (e.g., θ_0 or θ_1)

Assignment operator (update value)

Learning Rate (Step size). Controls how big a step we take.

Gradient (Slope). Direction of steepest ascent. We subtract it to go downhill.

How it Works

- If slope is +ve (right side): We subtract a positive amount → θ_1 decreases → moves left towards minimum.
- If slope is −ve (left side): We subtract a negative amount (which is adding) → θ_1 increases → moves right towards minimum.

Conclusion: By iteratively adjusting θ_0 and θ_1, we converge to the best-fit line parameters.

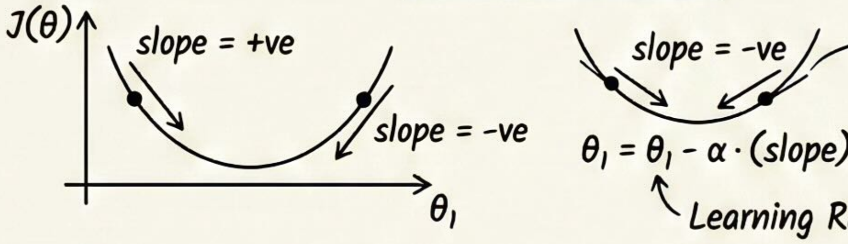
6. Gradient Descent & Evaluation

Gradient Descent (Optimization Algorithm)

Theory: An iterative algorithm to find the minimum of the cost function. Imagine walking down a hill (the cost function) in the steepest direction to reach the bottom (minimum cost).

$$\frac{\Delta J(\theta)}{\Delta \theta} = \frac{\partial J(\theta)}{\partial \theta} \longrightarrow \text{Gradient (Slope)} \rightarrow \text{Descent (Moving downwards) [cite: 11].}$$

Convergence Theorem: $\boxed{\theta_k = \theta_k - \alpha * \frac{\partial J(\theta)}{\partial \theta_k}}$ [cite: 11].



$\theta_1 = \theta_1 - \alpha \cdot (\text{slope})$

CT: $\theta_1 = \theta_1 - \alpha * (\text{slope})$
If slope is +ve, θ_1 decreases.
If slope is -ve, θ_1 increases [cite: 11].

Learning Rate (Step size) [cite: 11].

Visualizing Cost Function $J(\theta)$ (Example)

Data points: (1,2), (2,4), (3,6) [cite: 10].

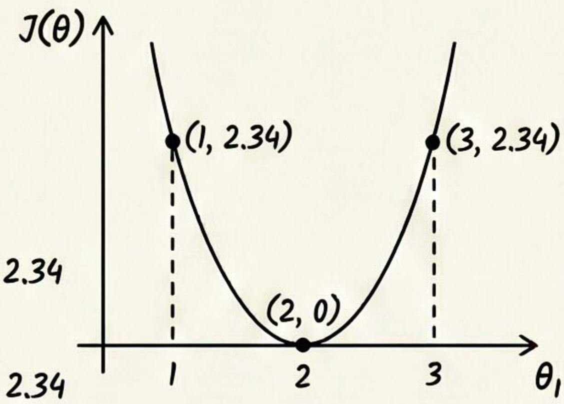
Hypothesis: $h_\theta(x) = \theta_0 + \theta_1 x$ [cite: 10]

Assuming $\theta_0 = 0$,

Case 1: $\theta_1 = 2$. $h_\theta(x) = 2x$. $J(\theta) = \frac{1}{6} \cdot (0^2+0^2+0^2) = 0$
Point (2, 0) [cite: 10].

Case 2: $\theta_1 = 1$. $h_\theta(x) = x$. $J(\theta) = \frac{1}{6} \cdot (1^2+2^2+3^2) = \frac{14}{6} = 2.34$
Point (1, 2.34) [cite: 10].

Case 3: $\theta_1 = 3$. $h_\theta(x) = 3x$. $J(\theta) = \frac{1}{6} \cdot (1^2+2^2+3^2) = \frac{14}{6} = 2.34$
Point (3, 2.34) [cite: 10].



Evaluation Metrics (Measuring Model Quality)

Theory: Metrics to quantify how well the model's predictions match the actual data [cite: 12].

- R-squared (R^2): $\boxed{1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2}}$ → $(\text{residual error})^2$ → variance [cite: 12].

  R^2 ranges from 0 to 1. 1 means perfect model, 0 means it explains nothing.
  E.g., 0.78 → 78% strong [cite: 12].

- Adjusted R-squared: $\boxed{1 - \frac{(1 - R^2)(n - 1)}{n - p - 1}}$ → n = no. of rows → p = no. of features → R^2 = R-squared [cite: 12].

Theory: Adjusted R^2 penalizes adding unnecessary features, preventing overfitting [cite: 12].